In general, the value of a polynomial $p(x)$ at $x=a$, denoted $p(a)$, is obtained by replacing x by a, where a is any real number.

Notice that the value of $p(x)$ can be zero for many possible values of x as in the second graph 3.8. So it is interesting to ask, for how many values of x, does p(x) become zero, and for which values ? We call these values of x, the zeros of the polynomial $p(x)$.

Once we see that the values of the polynomial are what we plot in the graph of the polynomial, it is also easy to notice that the polynomial becomes zero exactly when the graph intersects the X-axis.

The number of zeros depends on the line or curves intersecting x-axis.

For Fig. 3.7, Number of zeros is equal to 1

For Fig. 3.8, Number of zeros is equal to 2

Note

Number of zeros of a polynomial $\leq$ the degree of the polynomial

Value of a Polynomial

Value of a polynomial $p(x)$ at $x = a$ is $p(a)$ obtained on replacing x by a $(a \in R)$

For example,

Consider $f(x) = x^2 + 3x - 1$.

The value of $f(x)$ at $x = 2$ is

$f(2) = 2^2+3(2)-1 = 4+6-1 = 9.$

Zeros of Polynomial

(i) Consider the polynomial $p(x) = 4x^3 - 6x^2 + 3x - 14$

The value of $p(x)$ at $x = 1$ is $p(1) = 4(1)^3 - 6(1)^2 + 3(1) - 14$

$= 4 - 6 + 3 - 14$

$= -13$

Then, we say that the value of $p(x)$ at $x = 1$ is -13.

If we replace x by 0, we get $p(0) = 4(0)^3 - 6(0)^2 + 3(0) - 14$

$= 0 - 0 + 0 - 14$

$= -14$

we say that the value of $p(x)$ at $x = 0$ is -14.

The value of $p(x)$ at $x = 2$ is $p(2) = 4(2)^3 - 6(2)^2 + 3(2) - 14$

$= 32 - 24 + 6 - 14$

$= 0$

Since the value of $p(x)$ at $x = 2$ is zero, we can say that 2 is one of the zeros of $p(x)$ where $p(x) = 4x^3 - 6x^2 + 3x - 14$.

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